



Detect. Locate. Alert.



A LIGHTWEIGHT UAV-SAR SYSTEM FOR HUMAN DETECTION & MONOCULAR GEOLOCATION



Hamza Ghitri

University of Ain Temouchent Belhadj Bouchaib
Silesian University of Technology

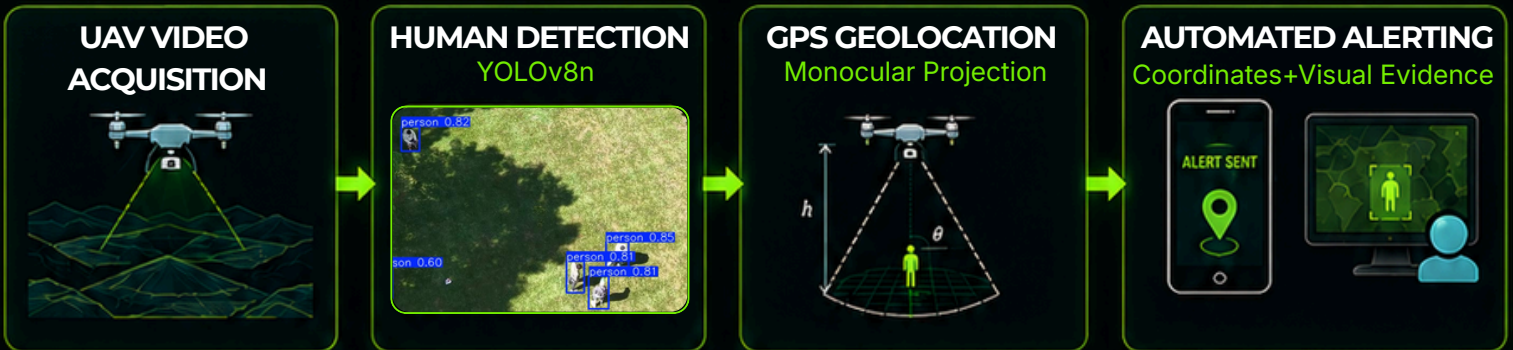
UAV-SAR DEPLOYMENT SYSTEM



Research Motivation

Search-and-rescue missions require rapid localization of missing persons across large and difficult environments. Although UAVs can quickly capture aerial imagery, current workflows often depend on manual video monitoring, which can delay response and increase operator workload. AlienSight addresses this gap by integrating lightweight human detection, monocular GPS geolocation, and automated alerting into a deployable UAV-SAR perception system.

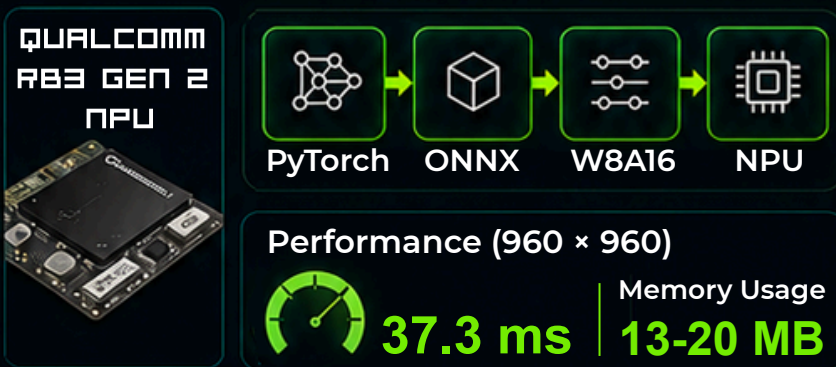
SYSTEM OVERVIEW



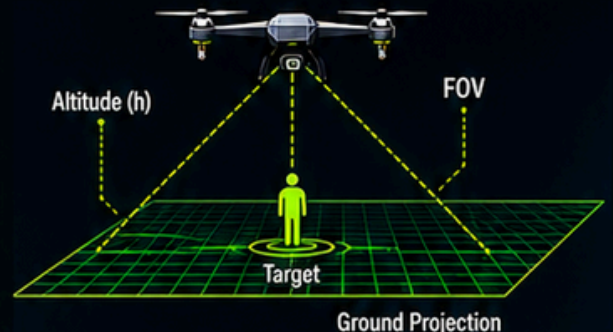
Proposed Framework

AlienSight is a lightweight end-to-end UAV-based search-and-rescue system that transforms aerial imagery into actionable rescue information. The framework processes UAV video and synchronized telemetry, detects humans using a YOLOv8n model adapted through VisDrone-to-HERIDAL transfer learning, estimates target GPS coordinates using a monocular geometric projection algorithm, and sends automated alerts containing coordinates, visual evidence, and navigation information to rescue operators.

EDGE DEPLOYMENT & OPTIMIZATION



GEOLOCATION ACCURACY



Mean Localization Error

1.01 m

(60 independent measurements)

Max Error: 1.86 m

Best Condition: 0.55 m

Research Identity

AlienSight translates UAV vision into actionable rescue intelligence. It focuses on lightweight aerial perception, practical deployment, and fast decision support for search-and-rescue operations. Built to reduce the gap between aerial observation and rescue action.



Real-Time
Detection



Precise
Geolocation



Edge
Optimization



Automated
Alerting



AlienSight Project Repository

Source code • Datasets • Publications • Resources



hamza.ghitri.edu@gmail.com



linkedin.com/in/hamza-ghitri



github.com/7amzaGH